

Network-based dissection of the SARS-CoV-2 host interactome: central complexes, a link-prediction test for missing components, drug repurposing, tissue and infection context, and pan-coronavirus conservation

Abstract

The affinity-purification–mass-spectrometry (AP-MS) map of SARS-CoV-2–human protein interactions ([Gordon et al. 2020](#)) identified 332 high-confidence host proteins engaged by 26 viral baits, but a map is a parts list rather than a mechanism. Here we reprocess the finished interactome from NDEx, integrate it with the human STRING physical network ([Szklarczyk et al. 2022](#)) and CORUM complex definitions ([Giurgiu et al. 2018](#)), and interrogate it with four questions under explicit, measurable standards. (i) Which host complex is central or hijacked? Community detection and centrality identify the **nuclear pore complex** as the most interconnected module — and a degree-preserving randomization confirms its cohesion exceeds degree-matched chance ($z = 67$, empirical $p < 0.001$), while multiple-testing-corrected over-representation analysis flags **mitochondrial protein import (TIM22)**, the **centrosome/SMYD2–PKA machinery**, and the **CUL2[^]ZYG11B ubiquitin ligase** as the most specifically enriched. (ii) Can we predict missing components? Adamic–Adar ([Adamic & Adar 2003](#)) and node2vec ([Grover & Leskovec 2016](#)) link prediction, scored on held-out STRING edges (AUROC 0.956–0.963) and shown to retain skill on cross-module edges (AUROC 0.817), nominated six nuclear-pore subunits absent from the map — and independent IntAct/BioGRID virus–host data corroborate all six as SARS-CoV-2 interactors, a genuine out-of-sample confirmation. (iii) Which targets are druggable? Mapping to DGIdb ([Freshour et al. 2020](#)), Open Targets ([Ochoa et al. 2020](#)) and ChEMBL ([Zdrazil et al. 2023](#)) classified 29 central targets by tractability. (iv) Which dependencies are conserved? Intersecting the SARS-CoV-2, SARS-CoV-1 and MERS host maps ([Gordon et al. 2020](#)) yielded 42 pan-coronavirus targets, contacted almost entirely by highly-conserved replication baits. We add three disease-context layers: infection transcriptomics ([Blanco-Melo et al. 2020](#)) show the hijacked complexes are commandeered post-transcriptionally (their mRNAs are largely unchanged during infection); tissue profiling ([GTEx Consortium 2020](#)) confirms the central modules are expressed in respiratory epithelium rather than being cell-line artifacts; and an L1000 connectivity test ([Subramanian et al. 2017](#)) finds the CK2 inhibitor silmitasertib does not reverse the infection signature transcriptomically; and a human-genetics cross-check finds none of the targets overlaps a COVID-19 Host Genetics Initiative risk gene, confirming the map captures physical dependencies rather than germline susceptibility. The therapeutic reading turns on a structural distinction: the most central and conserved machines are **structural scaffolds** (nucleoporins, TIM chaperones) with no approved chemistry, whereas the readily-repurposable approved drugs sit on **enzymatic bridge proteins** (HDAC2, RBX1, DNMT1), so the network's most important nodes and its most druggable nodes are different proteins, and closing that gap is the central medicinal-chemistry challenge these data pose. All findings are computationally-derived hypotheses that carry the detection bias of the underlying screen and require experimental validation.

1. Introduction

Viruses with small genomes are obligate hijackers of host machinery, and mapping the physical contacts between viral and host proteins has become a systematic route to understanding

pathogenesis and to nominating host-directed therapies. The approach was established for HIV, where AP-MS of viral baits against human cells revealed the host complexes the virus commandeers (Jäger et al. 2011), and was later generalized into comparative interaction mapping across related viruses to distinguish species-specific from conserved host dependencies (Shah et al. 2018). When SARS-CoV-2 emerged, the same pipeline produced a map of 332 high-confidence interactions between 26 viral baits and human proteins, and, alongside it, nominated existing drugs against those host factors (Gordon et al. 2020).

A physical interaction map is, by construction, an incomplete and biased sample of the true contact network. It is incomplete because AP-MS recovers only a fraction of true contacts, the "interactome is incomplete" property formalized by Menche et al. 2015, who showed that disease proteins occupy connected neighborhoods that are only partially observed; and it is biased because abundant, stable complexes are recovered more readily than transient ones. Two methodological consequences follow, and they motivate this study's design. First, interpreting a map is more informative when the map is embedded in a reference interactome and in curated complex and pathway resources, so that the sparse viral-contact data borrow statistical strength from dense human–human interaction data, in the manner of network medicine (Barabási et al. 2010). Second, the gaps are themselves informative: if a virus engages several subunits of a complex, the unobserved subunits are candidate missing components, and predicting them becomes a link-prediction problem on the human network (Lü & Zhou 2011).

This report treats the SARS-CoV-2 interactome as such an object. We do not rebuild it from raw spectra; we take the finished, peer-reviewed network and develop what it implies under a set of explicit standards: held-out scoring for every prediction, multiple-testing correction for every enrichment, degree-matched null models for every centrality claim, and source-level provenance for every value. We then extend the static topology in three directions that a physical map alone cannot address — whether the virus regulates its targets transcriptionally during infection, whether those targets are expressed in the cells the virus actually infects, and whether a nominated drug reverses the infection signature, and we ask which dependencies are shared with SARS-CoV-1 and MERS, since a host factor used by all three coronaviruses, bound through a conserved viral domain, is a more durable broad-spectrum target than one unique to a single virus or strain (Gordon et al. 2020).

2. Methods

2.1 Networks and reference resources

The finished SARS-CoV-2 host–pathogen interaction map (Gordon et al. 2020, Fig. 3) was retrieved from NDEx (network UUID 5d97a04a-6fab-11ea-bfdc-0ac135e8bacf; rights holder N. Krogan) as a CX2 stream and parsed with `ndex2`. The record contains 359 nodes and 481 drawn edges; classifying edges by their annotation separates 332 AP-MS bait–prey interactions (the high-confidence PPIs, each carrying a MiST score) from 149 CORUM-derived complex-membership edges added for visualization. Analyses of the viral–host contacts use the 332 AP-MS edges: 27 viral baits (the 26 viral proteins plus the Nsp5 C145A catalytic-dead control) and 332 human prey proteins. The map is bipartite and star-shaped — every prey is bound by exactly one bait, and no prey is shared between baits, which is why a reference human interactome is required for any neighborhood-based inference.

The human STRING v12.0 physical subnetwork (Szklarczyk et al. 2022) was downloaded, filtered to a combined score ≥ 700 , and reduced to an undirected symbol-level graph of 10,746 proteins and 86,519 interactions. Complex membership came from CORUM 5.3 (Giurgiu et al. 2018) (5,628 human complexes). Of the 332 host targets, 255 (77%) are present in the STRING reference and 162

are annotated in at least one CORUM complex. Integrating the AP-MS edges with the induced STRING subgraph gives the working graph on which community and centrality analyses were run.

2.2 Community detection, centrality and a degree-matched null

Community structure was computed by the Louvain method (Blondel et al. 2008) on the integrated graph (modularity $Q = 0.852$, 13 communities). Degree, betweenness and eigenvector centrality were computed on the largest connected component.

Because centrality scores are sensitive to node degree — a hub can score highly simply for being well-sampled, we tested the nuclear-pore complex against a **degree-preserving null**. We generated 1,000 randomized copies of the STRING physical network by double-edge-swap rewiring ($\approx 10 \times |E|$ swaps per network), which preserves every node's degree exactly (Maslov & Sneppen 2002). Eigenvector centrality is itself degree-confounded and therefore uninformative under this null (the eight nucleoporins' mean eigenvector centrality is, by construction, not elevated once degree is fixed); the discriminating statistic is the **within-set edge density** of the eight captured nucleoporins, the fraction of possible edges among them that are present, which measures cohesion independent of individual degree. The empirical p-value is the fraction of randomized networks whose within-set density meets or exceeds the observed value.

2.3 Enrichment

Over-representation of the 332 host targets was tested against CORUM complexes, Reactome pathways (Milacic et al. 2023) and Gene Ontology terms (Ashburner et al. 2000). For CORUM, the tested set was defined explicitly: each complex with at least two of its members among the 332 targets was tested ($k \geq 2$), against a hypergeometric background of the 5,150 genes that appear in at least one CORUM 5.3 human complex — this $k \geq 2$ filter is what defines the 87 complexes tested. Reactome used its UniProt-mapped human background (424 pathways with $k \geq 3$ tested); GO used `goatools` (Klopfenstein et al. 2018) with all GOA-annotated genes as background and `propagate_counts` enabled. Every family was Benjamini–Hochberg–corrected within itself; we report adjusted p-values (q).

2.4 Link prediction and missing-component nomination

Link prediction was run on the STRING physical network (the AP-MS map is degenerate for neighborhood methods). 15% of edges were held out (seed 42) with an equal number of sampled non-edges as negatives, verified absent from the full graph. Adamic–Adar scores (Adamic & Adar 2003) and node2vec embeddings (Grover & Leskovec 2016) (128-dimensional, followed by logistic regression on Hadamard features) were scored on the held-out set by AUROC, average precision and precision@k.

To test whether performance reflects genuine cross-module prediction rather than completion of dense annotated cliques, we **stratified** the held-out positives into within-module edges (both endpoints in the same Louvain community of the training graph) and between-module edges, and report metrics separately for each stratum. For the top hijacked complex, human proteins present in the CORUM reference complex but absent from the map were nominated as candidate missing components and ranked by aggregate Adamic–Adar and node2vec scores to the captured anchors; a leave-one-out test (hold out each captured subunit, rank it against the candidates plus random decoys) provided an internal recovery metric.

2.5 Drug mapping

The 29 most central and enriched targets were mapped to drugs and to tractability evidence via DGIdb 4.0 (Freshour et al. 2020), Open Targets (Ochoa et al. 2020) and ChEMBL (Zdravil et al. 2023). Interactions were separated into direct-mechanism types (inhibitor, antagonist, agonist, modulator) and indirect pharmacogenomic associations, and each target was classed Approved-drug / Clinical-stage / Tractable-novel / Hard-to-drug. Clinical-stage required either an Open Targets clinical-phase tractability tag or a named compound with a confirmed ChEMBL maximum phase ≥ 1 .

2.6 Cross-viral intersection

Human prey lists for SARS-CoV-2 (332), SARS-CoV-1 (365) and MERS (292) were parsed from their NDEx records (the SARS-CoV-1 and MERS comparative maps from Gordon et al. 2020, Science (Gordon et al. 2020); UUIDs `ba54a8e5-...` and `c14b148d-...`), uppercase-normalized, and intersected. The pan-coronavirus core is the exact three-way intersection.

2.7 Disease-context analyses

Infection directionality. Raw RNA-seq counts for GEO GSE147507 (Blanco-Melo et al. 2020) were analyzed with DESeq2 (`pydeseq2`) for infected-vs-mock contrasts in Calu-3 (Series 7, primary model) and A549-ACE2 (Series 6/16, supporting). Log2 fold changes were mapped onto the 332 host targets and summarized per module.

Tissue context. The 332 targets were profiled for expression in respiratory tissue using GTEx v8 median lung TPM (GTEx Consortium 2020), Human Protein Atlas single-cell distribution (Uhlén et al. 2015), and CELLxGENE CellGuide markers for alveolar type-2 (AT2) and ciliated cells. Targets were classed Expressed (≥ 10 TPM), Low (1–10) or Absent (< 1), and flagged in-vivo-plausible.

Transcriptomic drug reversal. A SARS-CoV-2 signature (top 150 up / 150 down genes by log2 fold change, $\text{padj} < 0.05$) from the Calu-3 DESeq2 result was queried against the L1000 chemical-perturbation library (693,215 signatures) via the login-free SigCom LINCS enrichment API (Evangelista et al. 2022) (Subramanian et al. 2017). Silmitasertib (CX-4945; BRD-K81458380) signatures were extracted and their signed connectivity z-scores compared to the library null (Wilcoxon and Mann–Whitney tests). The Broad clue.io Touchstone API was not used because it requires a registered key; L1000CDS2 (Duan et al. 2016) was queried as a cross-check.

Variant conservation. Each of the 42 pan-coronavirus core targets was assigned its SARS-CoV-2 bait from the AP-MS edges, and bait amino-acid conservation was assessed across six Omicron lineages (BA.1, BA.2, BA.5, XBB.1.5, JN.1, KP.3) using fixed substitutions (prevalence ≥ 0.5 vs Wuhan-Hu-1) from the CoV-Spectrum LAPIS open API (Chen et al. 2021); non-structural baits were mapped onto their ORF1a/ORF1b windows. Conservation density (substitutions per 100 aa, mean across lineages) classed each bait, with Spike as a positive control.

2.8 Reproducibility and provenance

Analyses ran in Python (`networkx`, `pandas`, `scikit-learn`, `gensim`, `goatools`, `statsmodels`, `pydeseq2`). Provenance is complete: every interaction traces to its NDEx source record, every enrichment to its CORUM/Reactome/GO identifier, every drug link to its database record, and every literature DOI resolves against CrossRef/OpenAlex. Data tables and figure-generating code are provided as artifacts.

3. Results

3.1 The map is a set of bait-specific modules, not a connected web

Displayed as a bait–prey incidence matrix (Figure 1), the high-confidence interactome is block-diagonal: each of the 332 human proteins is recovered by exactly one of the 27 viral baits, and no prey is shared between baits. This structure — a direct property of the AP-MS design, in which each bait is purified separately, means the map is best read as 27 independent target sets rather than one integrated network, and it is why every neighborhood-based inference in this study is performed on the human STRING reference rather than on the viral-contact edges themselves. The module blocks that later prove most interesting are already visible in the matrix: the nuclear-pore prey of Nsp9 and Orf6, the TIM22 import prey of Nsp4, and the CUL2 ligase prey of Orf10.

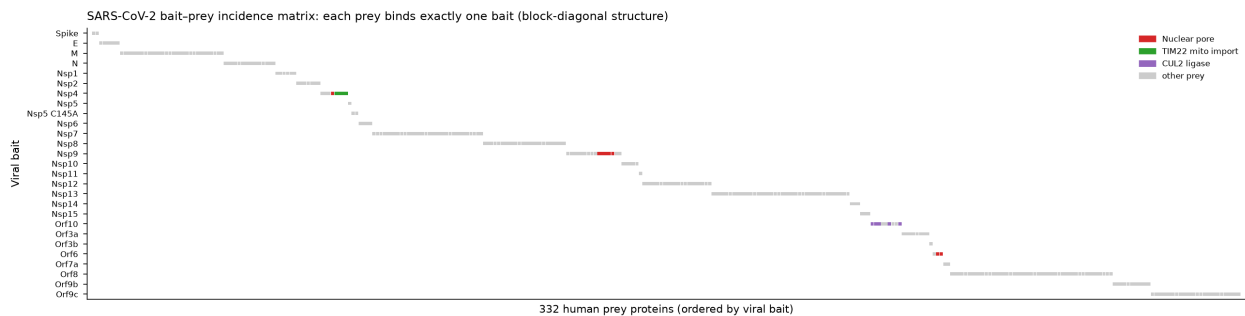


Figure 1. The interactome as a bait–prey incidence matrix. Rows are the 27 viral baits (ordered structural → Nsp → Orf); columns are the 332 human prey proteins, ordered so that each bait's targets form a contiguous block. A colored square marks each of the 332 AP-MS interactions. The block-diagonal pattern shows that every prey binds exactly one bait. Colored blocks highlight the three modules that recur throughout this study: nuclear pore (red), TIM22 mitochondrial import (green) and the CUL2 ubiquitin ligase (purple).

3.2 Two complementary readouts of "central" and "hijacked"

Community detection resolved 13 modules (modularity $Q = 0.852$), most of which correspond to a single bait's target set annotated to a known complex. Two distinct questions — which module is most *connected*, and which is most *specifically enriched* — have two defensible but different answers.

By connectivity, the **nuclear pore complex** dominates. Its captured nucleoporins (NUP98, NUP214, NUP88, NUP62, NUP54, NUP210, NUP58 and RAE1) hold the top degree and eigenvector-centrality ranks in the integrated graph (Figure 2), consistent with SARS-CoV-2's documented interference with nucleocytoplasmic transport through Orf6 and Nsp9. Because raw centrality can reward well-sampled hubs, we tested the pore against a degree-preserving null (Figure 3): across 1,000 degree-matched rewirings, the within-set edge density of the eight nucleoporins is 0.93 observed versus a null mean of 0.005 ($z = 67$, far above the null, empirical $p < 0.001$). The nucleoporins are therefore far more cohesively interconnected than their degrees alone would predict — the pore's centrality is a real structural signal, not a sampling artifact. (As expected, mean eigenvector centrality is not elevated under this null, since it is itself a function of degree; density is the metric that isolates cohesion from degree.)

Central human proteins in the SARS-CoV-2 interactome (bar colour = complex module)

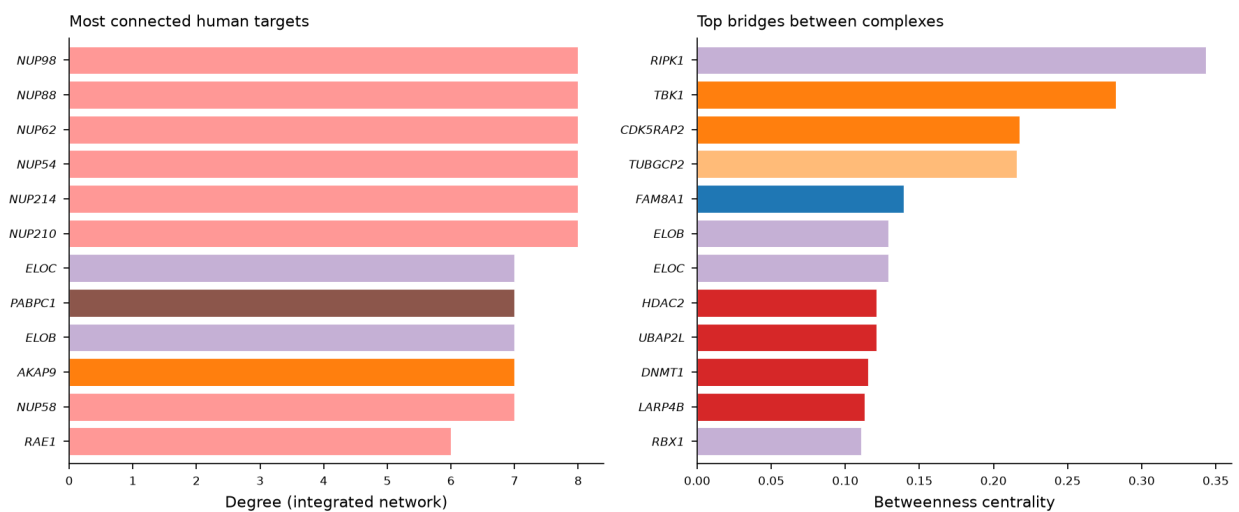


Figure 2. Most central human proteins. Top-ranked targets by degree (left) and betweenness (right) in the integrated network, colored by community. Nucleoporins hold the top degree ranks; signaling bridges (RIPK1, TBK1) and centrosomal proteins (CDK5RAP2, TUBGCP2) hold the top betweenness ranks.

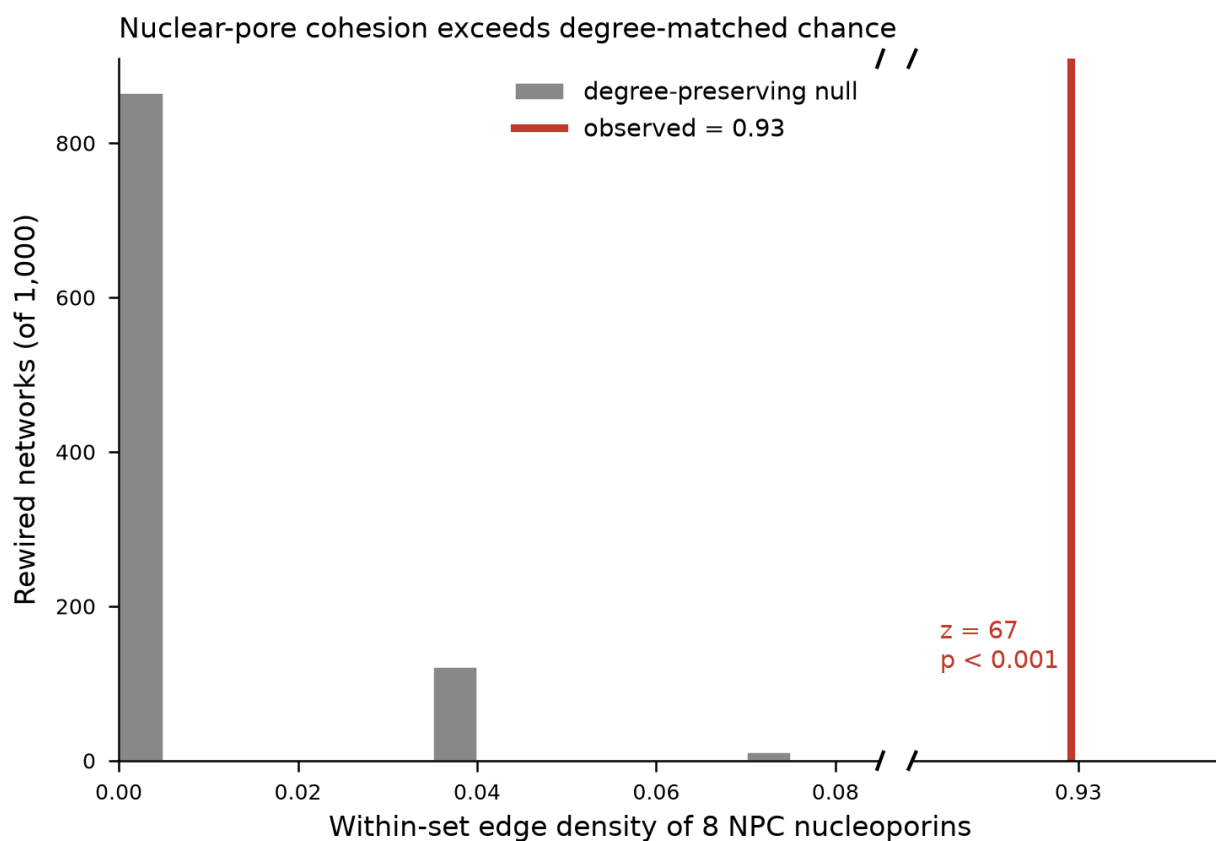


Figure 3. Nuclear-pore cohesion exceeds degree-matched chance. Distribution of the within-set edge density of the eight captured nucleoporins across 1,000 degree-preserving (double-edge-swap) rewirings of the STRING physical network (grey); the observed value (red, 0.93) lies far outside the null ($z = 67$, $p < 0.001$). Note the broken x-axis.

By enrichment specificity, a different set of machines rises to the top. Of 87 CORUM complexes tested, 80 were significant after correction; of 424 Reactome pathways, 170. Once broad terms (e.g. "protein binding") are set aside, the most specifically over-represented machinery is **mitochondrial protein import (TIM22)** — the strongest specific Reactome pathway and among the strongest CORUM and GO terms (23–78× fold enrichment), driven by Nsp4, together with the **centrosome / SMYLE–PKA** machinery (Nsp13) and the **CUL2^ZYG11B ubiquitin ligase** (Orf10) (Figure 4). The distinction is worth stating plainly: centrality flags the nuclear pore because its subunits are densely intertwined in the human interactome, whereas enrichment flags TIM22 because the virus captures an unusually complete slice of a small, specific complex. Both are correct readings of different questions.

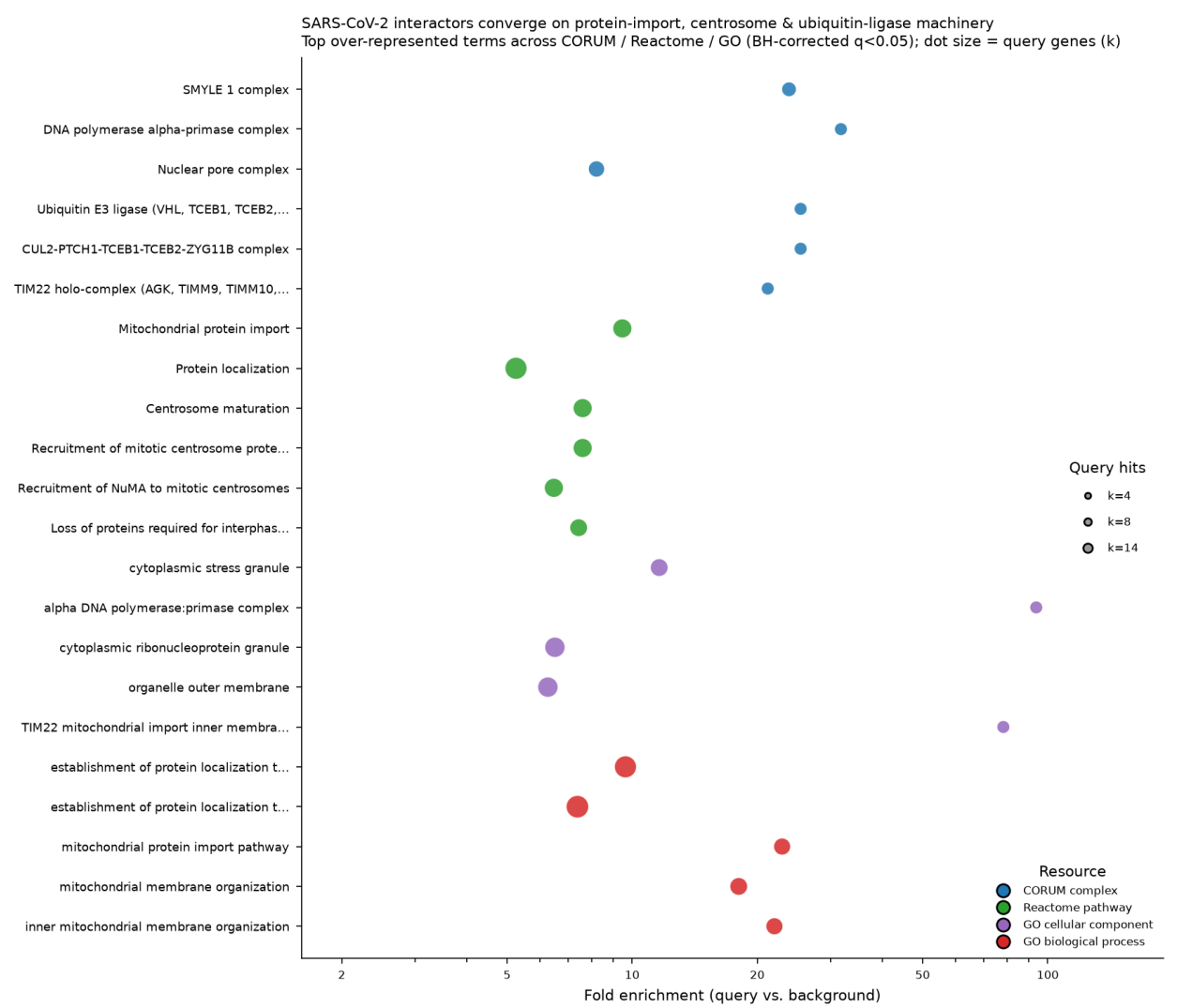


Figure 4. Complexes and pathways over-represented among the host targets. Top specifically-enriched terms from CORUM, Gene Ontology and Reactome (Benjamini–Hochberg q < 0.05), by fold enrichment. Mitochondrial protein import, centrosome/spindle and ubiquitin-ligase terms recur across resources.

3.3 Link prediction recovers missing nuclear-pore subunits, and works across modules

Held-out link prediction on the STRING physical network performed well: Adamic–Adar reached AUROC 0.956 and node2vec 0.963 on 15% held-out edges. A high AUROC on a network with dense complexes can, however, be inflated by "clique completion" — recovering the last missing edge inside an already-dense module is easy and biologically shallow. We therefore stratified the held-out edges

(Figure 5). Within-module edges are indeed easier (AUROC 0.977), but performance on **between-module** edges, genuine cross-complex prediction, remains well above chance (AUROC 0.817, average precision 0.687). The predictor is not merely closing triangles inside complexes; it retains real skill on edges that bridge distinct modules.

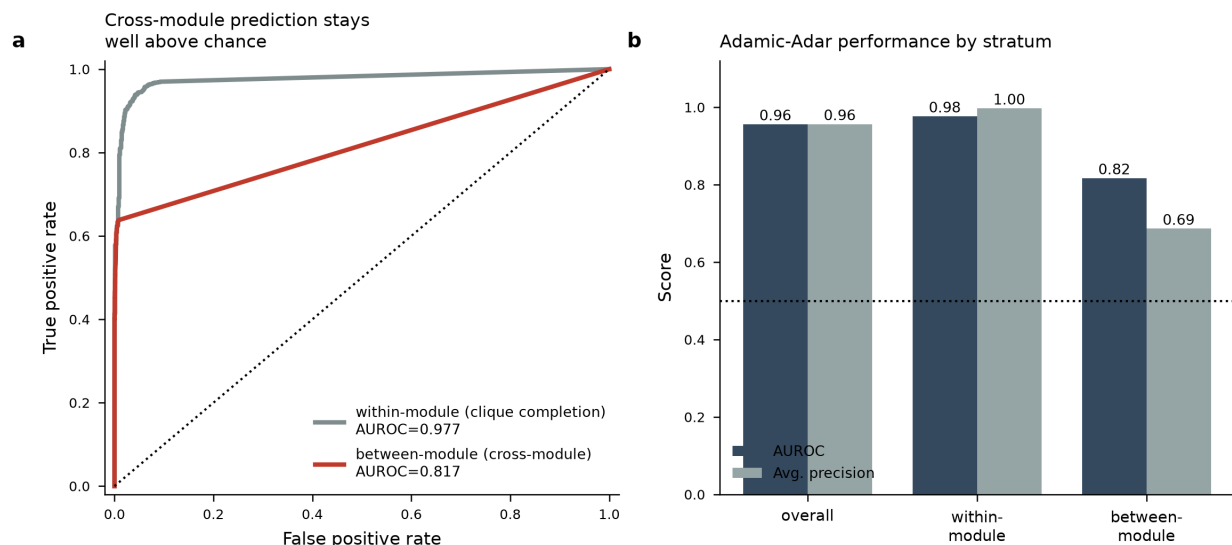


Figure 5. Cross-module prediction stays above chance. ROC curves (left) and metric bars (right) for Adamic–Adar link prediction on held-out STRING edges, stratified into within-module (clique-completion) and between-module (cross-module) edges. Both strata exceed chance; the between-module AUROC of 0.817 shows the signal is not only clique completion.

Applied to the top hijacked complex, this machinery nominates missing components. The nuclear pore complex (CORUM:8674) has 31 subunits, of which 8 are captured in the map and 23 are absent; all 23 absent subunits are present in STRING. Ranked by aggregate Adamic–Adar and node2vec score to the eight captured anchors, the top nominations are **NUP205**, **RANBP2**, **NUP133**, **NDC1**, **NUP155** and **NUP188** (Figure 6). A leave-one-out control — holding out each captured nucleoporin and ranking it against the 23 candidates plus 200 random decoys, recovered all eight within the top 20 (recovery AUROC 1.0). This control confirms the ranking is internally consistent, but it is not independent evidence: for a fully-annotated complex like the nuclear pore, recovering additional subunits from CORUM co-membership is close to tautological, so the recovery AUROC of 1.0 should be read as an internal-consistency check rather than as proof the virus engages these proteins. More generally, these are predictions of co-membership plausibility in the human interactome — which proteins most resemble the captured subunits — not predictions of direct viral contact. Genuinely independent support must come from experimental data not used to make the prediction; the next section tests the six nominations against out-of-sample SARS-CoV-2 interaction evidence, and they remain hypotheses for co-immunoprecipitation or proximity-labeling validation.

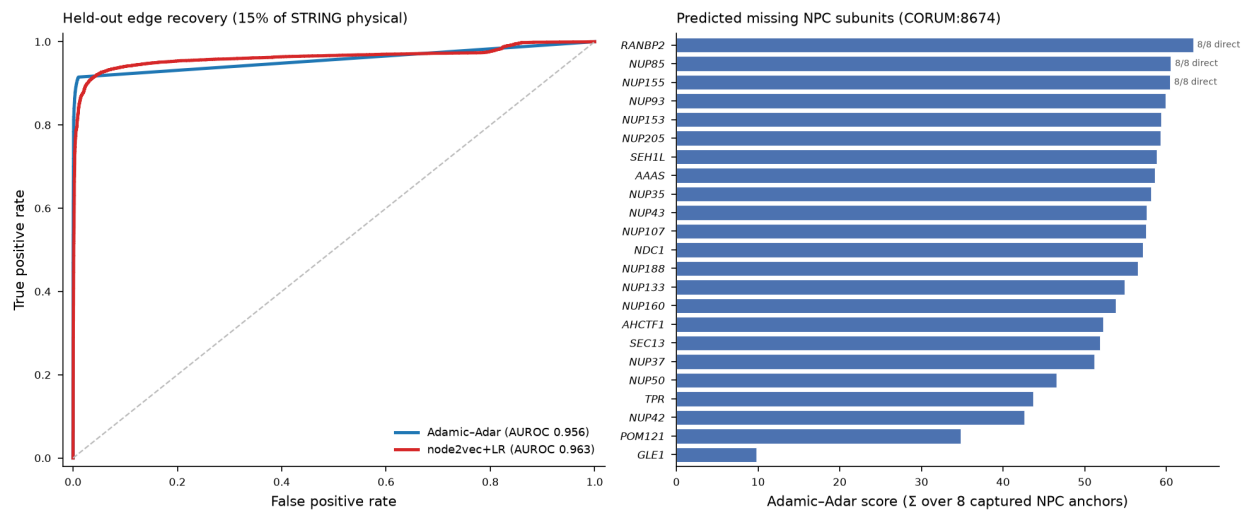


Figure 6. Predicted missing nuclear-pore components. Absent nuclear-pore subunits ranked by aggregate Adamic–Adar score to the eight map-captured anchors; annotations give the number of direct STRING edges to anchors. NUP205, RANBP2 and NUP133 top the ranking.

3.4 Independent virus–host data corroborate all six predicted nucleoporins

A missing-component prediction is settled only by data that were not used to make it. The six nominations were compared against SARS-CoV-2 virus–host interaction datasets generated separately from the Gordon et al. 2020 map, curated in IntAct and the BioGRID SARS-CoV-2 network, with any record deriving from the original map excluded. **All six nominated nucleoporins are independently corroborated** (Figure 7). Each is reported as an interactor of one or more SARS-CoV-2 viral proteins in post-2020 interactome studies: NUP155 across eight independent studies and eight viral partners, NUP205, NUP133, NDC1 and NUP188 across five studies each, and RANBP2 across three. Critically, three of the six are reported bound to the very baits that captured the mapped anchors: RANBP2 and NUP188 to Orf6 (Lee et al. 2021), the bait that captures NUP98 and RAE1, and NUP155 to both Orf6 and Nsp4 (Liu et al. 2021) (Broderick et al. 2025), Nsp4 being the bait that captured NUP210. The corroborating interactions are high-throughput co-complex and proximity associations (IntAct annotates them as `interaction_type = association`; the BioGRID edges are affinity-capture- or proximity-labeling-derived) rather than proven direct binary contacts, so they establish co-membership in the virally engaged nuclear pore rather than a direct viral contact. That distinction matters: the prediction was made purely from human co-membership structure, and it is corroborated by independent evidence of the same co-membership under viral engagement — a genuine out-of-sample confirmation that nonetheless leaves the direct Orf6/Nsp4/Nsp9 contact to be established by targeted co-immunoprecipitation. The fifteen supporting studies, each with a CrossRef-resolvable DOI, are listed in the accompanying evidence table.

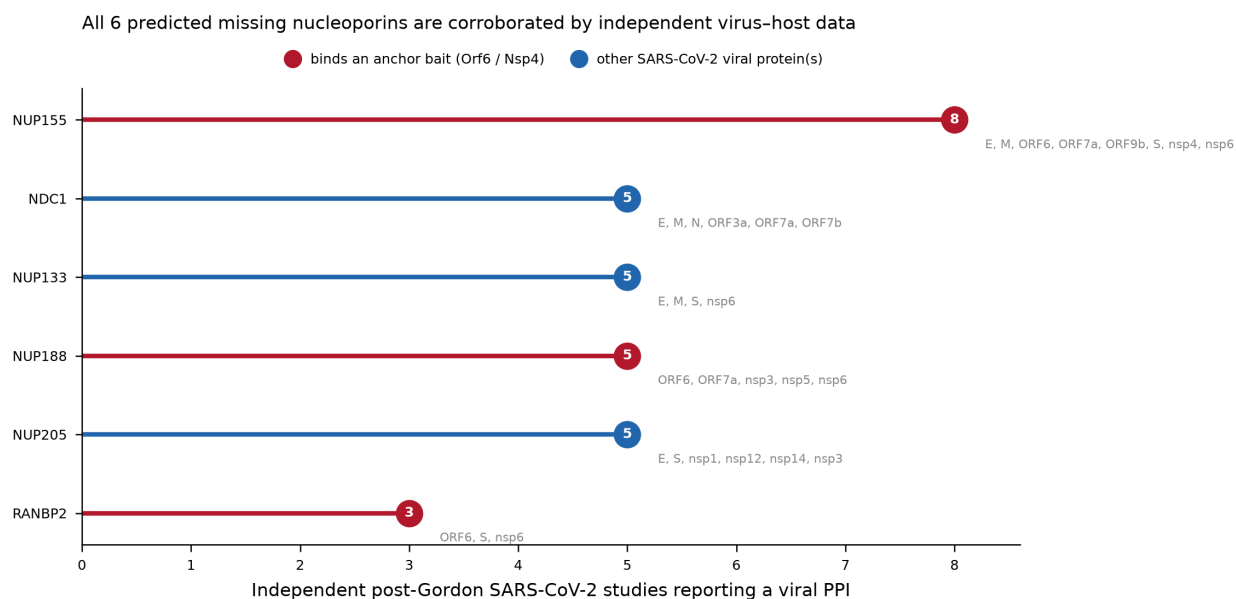


Figure 7. All six predicted nucleoporins are corroborated by independent virus–host data. Number of post-Gordon SARS-CoV-2 interactome studies reporting a viral interaction for each nominee; red marks nominees reported bound to an anchor bait (Orf6 or Nsp4), the baits that captured the eight mapped nucleoporins. Grey text lists the viral partners reported for each.

3.5 The readily-druggable targets are enzymatic bridges, not scaffolds

Mapping the 29 central and enriched targets to DGIdb, Open Targets and ChEMBL produced a clear split (Figure 8). The proteins with approved, direct-mechanism drugs are enzymes that sit as bridges between modules rather than the central scaffolds themselves: **HDAC2** (vorinostat, romidepsin, belinostat, panobinostat, givinostat), **DNMT1** (azacitidine, decitabine) and **RBX1** (the immunomodulatory imides lenalidomide, pomalidomide, thalidomide, which act through the CUL2/CRL machinery). TBK1 (amlexanox) and RIPK1 (ponatinib, off-target) add two more. In contrast, the most central and most conserved machines — the nucleoporins and the TIM22 chaperones, are structural scaffolds with, at best, low-confidence antibody-based tractability and no small-molecule chemistry, and they are intracellular, placing them out of reach of antibody modalities. The therapeutic implication is not that HDAC2 or DNMT1 are good COVID-19 drugs, they are cytotoxic oncology agents (see §4.4), but that the network's importance and its druggability are carried by different proteins.

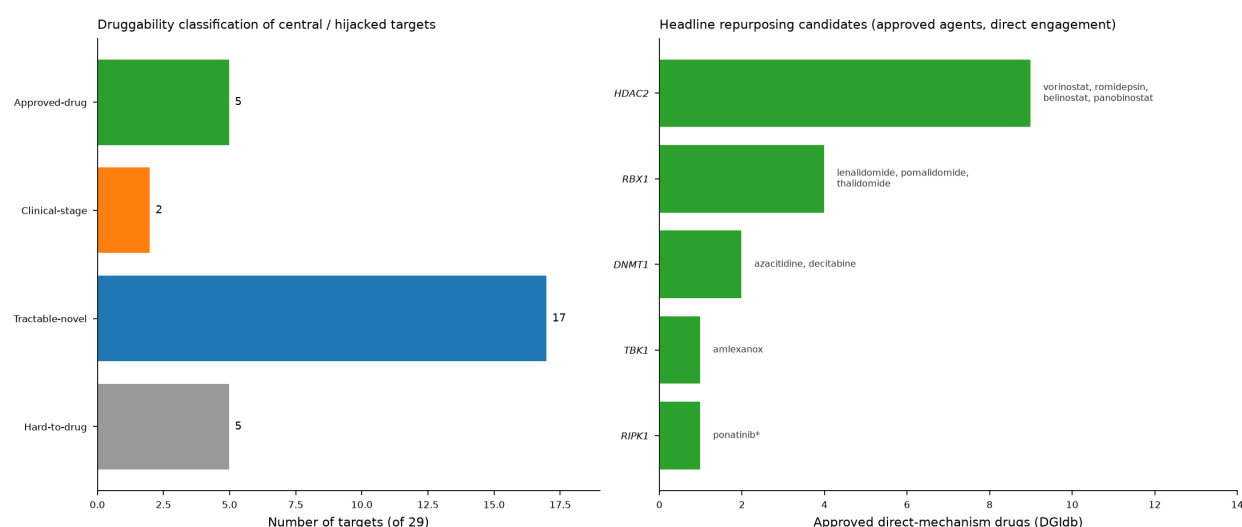
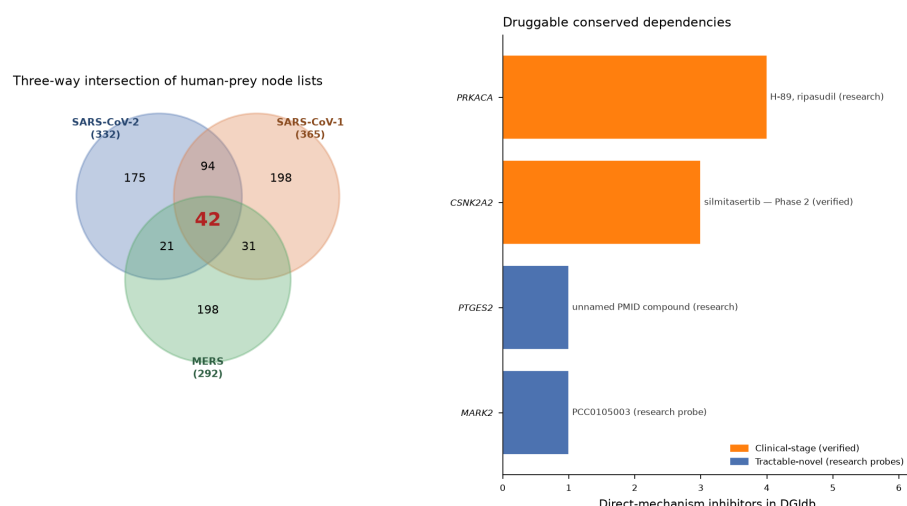


Figure 8. Tractability classification of 29 central targets. Each target classed Approved-drug / Clinical-stage / Tractable-novel / Hard-to-drug from DGIdb, Open Targets and ChEMBL evidence. Approved direct-mechanism drugs concentrate on enzymatic bridge proteins (HDAC2, RBX1, DNMT1), not on the central nuclear-pore or TIM22 scaffolds.

3.6 Forty-two host factors are shared across three coronaviruses

Intersecting the human prey lists of SARS-CoV-2 (332), SARS-CoV-1 (365) and MERS (292) yields a pan-coronavirus core of **42 proteins** targeted by all three (the pairwise and three-way regions sum exactly to the union of 759, confirming the set operation). Among these 42, small-molecule-tractable candidates with clinical-phase evidence are limited to two — **CSNK2A2** (casein kinase II α' , inhibited by silmitasertib, ChEMBL maximum phase 2) and **PRKACA** (PKA catalytic α), with the remainder tractable-novel or hard-to-drug (Figure 9). CK2 has independent literature support as an antiviral and cancer target (Borgo et al. 2021), making it the strongest broad-spectrum hypothesis from the topology alone. That headline carries a caveat that the conservation analysis (§3.10) makes explicit: CSNK2A2 is engaged by the nucleocapsid (N) protein, one of the more variable baits, whereas the large block of centrosome and PKA targets sits behind the highly-conserved Nsp13 helicase — so the most druggable core node is not the one bound through the most durable viral domain. Two orthogonal tests of the CK2 hypothesis follow: infection transcriptomics (§3.9) and out-of-sample durability (§3.10).



No conserved-core protein has an approved direct-mechanism drug. Only PRKACA & CSNK2A2 have verified clinical-stage compounds.

Figure 9. The 42-protein pan-coronavirus core and its druggability. Host proteins targeted by SARS-CoV-2, SARS-CoV-1 and MERS, with tractability class. Two (CSNK2A2, PRKACA) reach clinical-stage small-molecule tractability; most are structural or scaffolding proteins without approved chemistry.

3.7 The central modules are expressed in respiratory epithelium, not cell-line artifacts

Because the AP-MS map was generated in HEK293T cells, a central concern is whether the hijacked complexes are relevant to the cells SARS-CoV-2 actually infects. Profiling the 332 targets against GTEx v8 lung expression, Human Protein Atlas single-cell distribution and CELLxGENE markers for alveolar type-2 and ciliated cells (Figure 10) shows the concern is largely unfounded for the modules that matter: 282/332 targets (85%) are expressed in lung (≥ 10 TPM) and 288/332 are flagged in-vivo-plausible. Critically, **all 17 nuclear-pore, TIM22 and CUL2-ligase genes are lung-expressed** and detected in AT2 and ciliated cells — these are housekeeping machines, and the network's central and enriched modules are genuine respiratory-epithelium dependencies rather than 293T-specific interactions. Among the 42-gene core, only MIPOL1 falls below the expression threshold, a single weak candidate that should be de-prioritized.

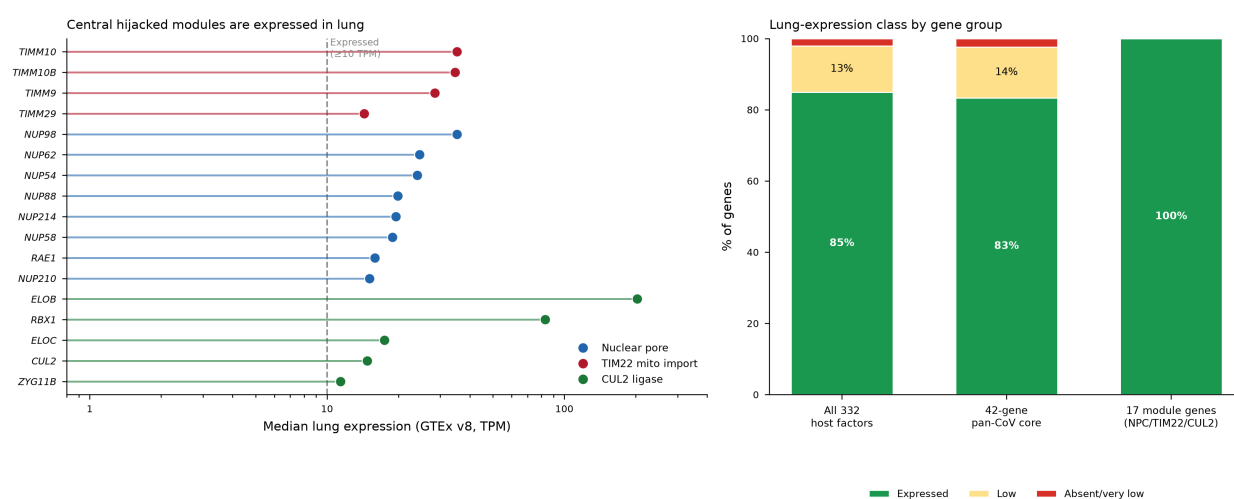


Figure 10. The central hijacked modules are expressed in lung. Left: median GTEx v8 lung expression of the 17 nuclear-pore, TIM22 and CUL2-ligase genes, all above the 10-TPM expressed

threshold. Right: lung-expression class for all 332 targets, the 42-gene core, and the 17 module genes; the module genes are 100% expressed.

3.8 The virus hijacks its targets post-transcriptionally

A physical map cannot say whether the virus also regulates its targets' expression. Mapping infected-vs-mock differential expression from GSE147507 (Calu-3, DESeq2) onto the 332 targets (Figure 11) shows that it largely does not: 216 of 332 targets are transcriptionally unchanged during infection, and the three central modules sit squarely within the normal transcriptome band. Of the eight nuclear-pore subunits, seven are unchanged; only NUP210 is significantly down-regulated ($\log_2$ fold change -0.67 , $q = 1.9 \times 10^{-7}$). The interpretation is mechanistically clean: SARS-CoV-2 commandeers the nuclear pore, TIM22 import and the CUL2 ligase **post-transcriptionally** — by physical engagement and, as the phosphoproteome shows (Bouhaddou et al. 2020), by post-translational modification, rather than by altering the abundance of their mRNAs. This is consistent with a virus that repurposes existing host machinery on the timescale of a single infection cycle rather than rewiring host transcription.

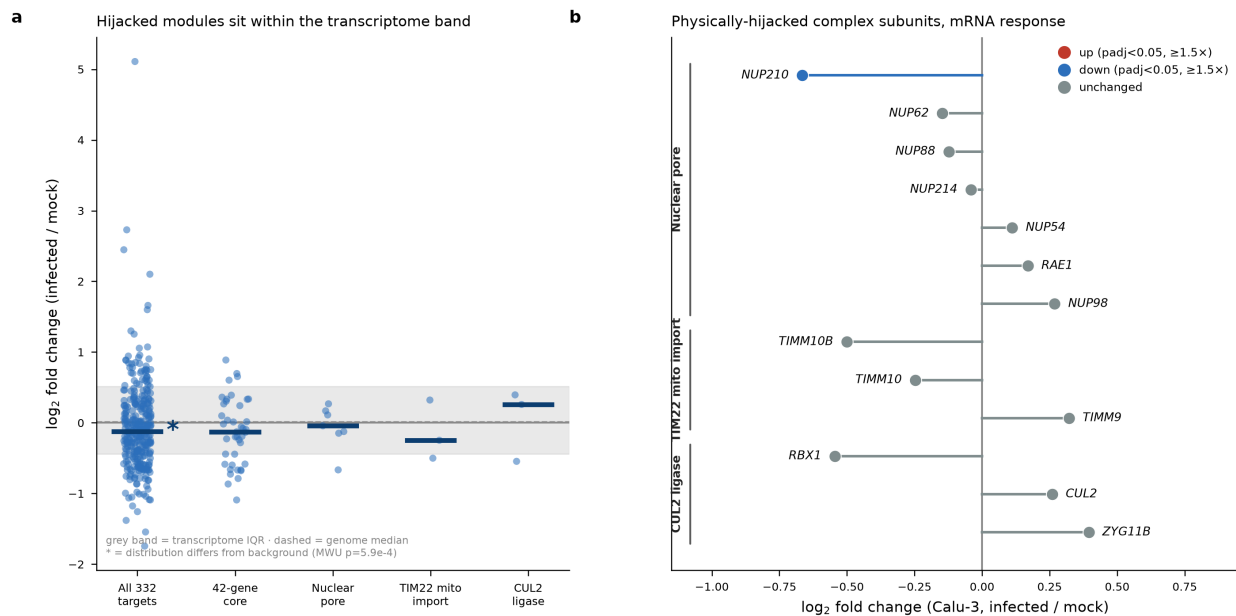


Figure 11. Hijacked complexes are transcriptionally unchanged during infection. Left: distribution of infected-vs-mock $\log_2$ fold change (Calu-3, GSE147507) for all 332 targets, the 42-gene core, and the three central modules; all hijacked modules lie within the transcriptome interquartile band. Right: per-subunit fold change for the physically-hijacked complexes; only NUP210 crosses the significance threshold.

3.9 The CK2 inhibitor does not reverse the infection signature transcriptomically

The topology nominates CSNK2A2/CK2 (silmitasertib) as the strongest broad-spectrum candidate (§3.6). We tested this orthogonally by asking whether the silmitasertib transcriptomic signature reverses the SARS-CoV-2 infection signature, using real L1000 data. Against the full chemical-perturbation library (693,215 signatures) via SigCom LINCS, the 353 silmitasertib signatures show **no net reversal**: mean connectivity $z = +0.05$ (Wilcoxon $p = 0.74$; Mann–Whitney vs the library null $p = 0.88$) (Figure 12). This is a genuine negative result, and it refines rather than refutes the CK2 hypothesis. A host-factor inhibitor is not expected a priori to reverse the bulk infection transcriptome — its rationale is mechanistic (blocking a hijacked kinase) rather than transcriptomic, and the infection signature here is dominated by interferon signaling in one cell line at one timepoint. CK2

therefore remains a topology-supported broad-spectrum hypothesis with independent literature backing, but it does not gain transcriptomic-reversal support, and the strength of the candidate should be stated at that level.

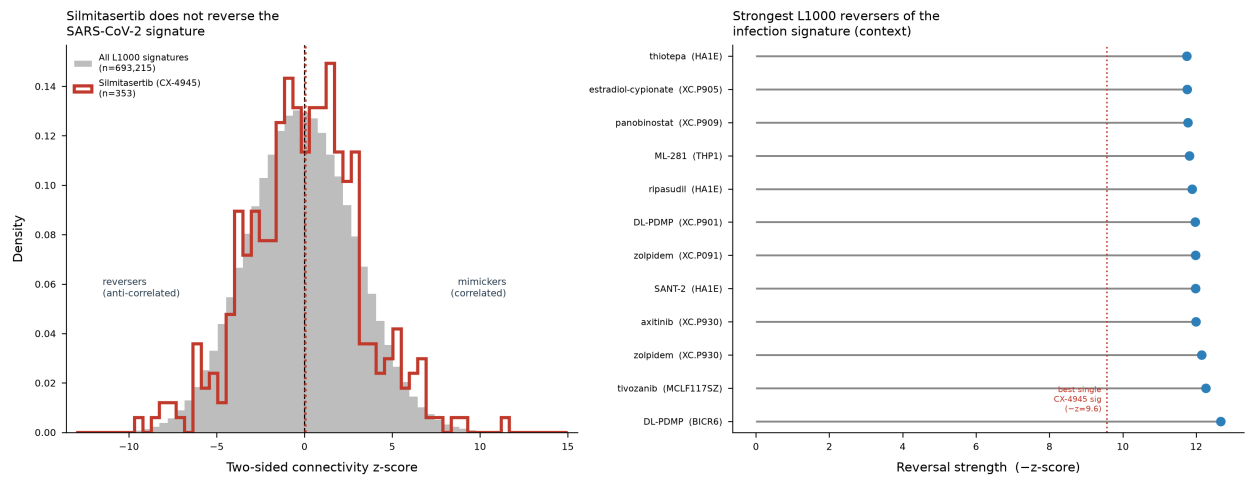


Figure 12. Silmitasertib does not reverse the infection signature. Connectivity z-scores of the 353 silmitasertib (CX-4945) L1000 signatures against the SARS-CoV-2 Calu-3 signature, relative to the 693,215-signature library null. The distribution is centered near zero (mean $z = +0.05$), indicating no systematic reversal.

3.10 The conserved targets are bound through conserved viral domains

A broad-spectrum target is most durable if the viral protein that engages it is itself evolutionarily stable. Assigning each of the 42 pan-coronavirus core targets to its SARS-CoV-2 bait shows they are contacted almost entirely by the **viral replication machinery**: Nsp13 (helicase) binds 15 of them, Nsp8 binds 11, and Nsp7 binds 7 — 33 of 42 through three replication-complex proteins, and none is bound by Spike. Scoring bait conservation across six Omicron lineages against a Spike positive control (Figure 13), these replication baits are strongly conserved: Nsp7 and Nsp8 carry zero fixed amino-acid substitutions in any lineage, and Nsp13 carries 0.17 substitutions per 100 aa, versus 3.04 per 100 aa for Spike, an 18-fold difference. 37 of the 42 core targets are therefore engaged through highly-conserved viral domains and flagged as the most durable therapeutic targets; the five exceptions are bound by more variable baits (NUP210 by Nsp4; CSNK2A2 and RRP9 by N; MARK2 by Orf9b; TUBGCP2 by M). Notably, the top broad-spectrum drug target CSNK2A2 is bound by the moderately-variable N protein, whereas the 15 Nsp13-bound centrosome/PKA targets sit behind a highly-conserved helicase.

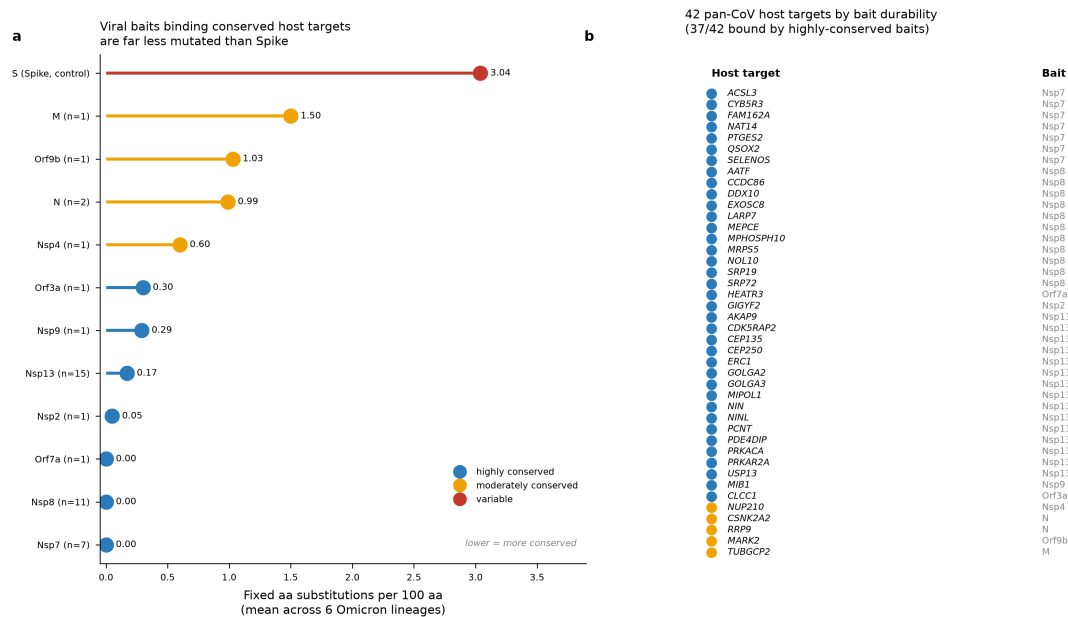


Figure 13. Conserved host targets are bound through conserved viral baits. Left: fixed amino-acid substitutions per 100 aa (mean across six Omicron lineages) for each viral bait binding the 42 core targets, with Spike as a positive control; replication baits (Nsp7/8/13) are far less mutated. Right: the 42 core targets with their baits, colored by bait durability; 37/42 are bound by highly-conserved baits.

3.11 The physical host factors do not overlap COVID-19 germline risk genes

Physical host factors and germline susceptibility genes capture different biology: a protein the virus grabs during infection need not sit at a locus where human sequence variation alters COVID-19 risk, and testing our targets against human genetics makes that separation explicit. Intersecting the union of the 29 central and 42 core targets (65 unique genes) with COVID-19 genetic associations in the Open Targets Platform (release 25.06, disease MONDO_0100096) and with the COVID-19 Host Genetics Initiative (HGI) meta-analysis (COVID-19 Host Genetics Initiative 2021) shows the overlap is essentially null (Figure 14). **None** of the 65 targets coincides with a known HGI genome-wide-significant gene (OAS1/2/3, IFNAR2, TYK2, DPP9, LZTFL1/SLC6A20, FOXP4, ABO, MUC5B). Forty targets carry a non-zero Open Targets COVID-19 score, but these are driven by the affected-pathway datatype (the Reactome nuclear-transport pathway assigns a uniform pathway score to the eleven captured nucleoporins) and by literature text-mining, not by germline genetics. Only five targets carry any genetic-association score, and all five trace to a single Polish whole-genome-sequencing cohort study (Słomian et al. 2023, PMID 36662838) rather than to the HGI meta-analysis; of these, credible-set fine-mapping supports the target as the likely causal gene for only three (NUP210, PDE4DIP, NOL10), and NUP210's locus (3p24.2) is distinct from the well-known 3p21.31 severity locus. This null overlap is the expected result and a useful negative control: it confirms the interactome analysis is identifying physical dependencies rather than re-discovering susceptibility loci, and it means the two lines of evidence are complementary rather than redundant.

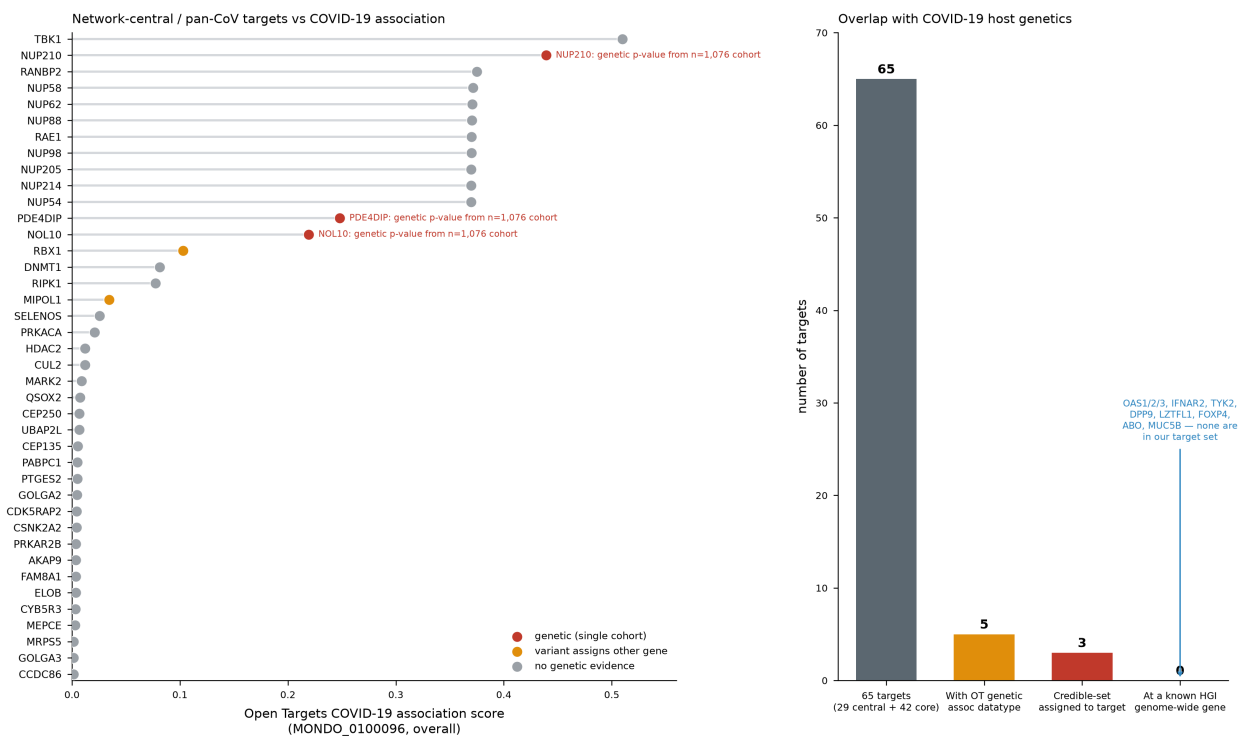


Figure 14. Physical host factors show minimal overlap with COVID-19 germline risk genes. Left: Open Targets COVID-19 association score for the 65 central and core targets, colored by genetic-evidence status. Right: of 65 targets, 5 carry an Open Targets genetic-association datatype score (all from one cohort), 3 are the credible-set causal gene, and 0 coincide with a known HGI genome-wide gene.

4. Discussion

4.1 Two answers to "which complex is doing the work," and why both matter

The central methodological result of this study is that "central" and "hijacked" are not one question. By interconnectivity in the human interactome, the nuclear pore complex is the standout, and a degree-matched null confirms its cohesion is real rather than a sampling artifact (§3.2, Figure 3). By enrichment specificity, mitochondrial protein import (TIM22), the centrosome/SMYD2–PKA machinery, and the CUL2[^]ZYG11B ligase stand out. These are complementary, not contradictory: the pore is central because nucleoporins are densely interwired, while TIM22 is enriched because the virus captures a large fraction of a small complex. Both readings have independent biological support — SARS-CoV-2 disrupts nucleocytoplasmic transport through Orf6 and Nsp9 (Miorin et al. 2020) (Lee et al. 2021), and Orf6 in particular antagonizes interferon signaling by sequestering the NUP98–RAE1 complex (Lei et al. 2020). Collapsing these into a single answer to "which complex is doing the work" would obscure a real distinction between two different notions of importance.

4.2 Predicted missing components are co-membership hypotheses

The link-prediction framework recovers biologically sensible missing nucleoporins, the stratified analysis (§3.3) shows the signal is not merely clique completion, and — most importantly — independent virus–host data corroborate all six nominations (§3.4). That external check is what elevates the predictions above a within-CORUM tautology: the leave-one-out control confirms only

internal consistency, whereas the IntAct and BioGRID evidence was generated separately from the map and still reports each nominee as a viral interactor, three of them through the same Orf6/Nsp4 baits that engage the mapped anchors. The interpretation remains bounded. The corroborating data are co-complex and proximity associations, not proven direct binary contacts, so whether the virus engages NUP205 or RANBP2 through a direct interface is still an experimental question for co-immunoprecipitation or proximity labeling. The value of the prediction is prioritization with independent support: of the 23 absent nucleoporins, the model names the handful most worth testing first, and the literature already places those same proteins at the viral interface.

4.3 Importance and druggability are carried by different proteins

The therapeutic geometry of this interactome is its most actionable feature. The proteins that are most central (nucleoporins), most specifically enriched (TIM22 chaperones) and most conserved across coronaviruses (§3.10) are structural scaffolds — intracellular, largely enzyme-free, and without approved small-molecule chemistry. The proteins with approved, direct-mechanism drugs (HDAC2, DNMT1, RBX1) are enzymatic bridges at the periphery of the central modules. This gap, between where the network's importance sits and where the existing pharmacology sits, is the central medicinal-chemistry problem these data pose, and it points toward modalities that can drug scaffolds: molecular-glue and PROTAC degraders that act through the very CUL2/RBX1 machinery the virus itself exploits (Békés et al. 2022), and which do not require a classical enzymatic pocket.

4.4 Repurposing candidates are starting points, and some carry real feasibility caveats

The approved drugs surfaced here are targets' drugs, not the disease's drugs, and two feasibility caveats must be stated explicitly. HDAC2 and DNMT1 are the best-precedented targets by tractability, but their approved inhibitors — vorinostat, romidepsin, panobinostat, azacitidine, decitabine, are cytotoxic epigenetic agents approved for hematological malignancies, with narrow therapeutic windows and substantial toxicity. Deploying them against an acute, mostly self-limiting respiratory infection is difficult to justify on a risk–benefit basis, and their appearance here reflects the density of oncology drug development around chromatin enzymes rather than a specific antiviral rationale. RBX1's link is more mechanistically interesting: the immunomodulatory imides act through the CUL–RING ligase system that Orf10 engages, connecting an approved drug class to a hijacked complex, though the direction of effect on viral replication is untested. The CK2 candidate (CSNK2A2/silmitasertib) is the most attractive on paper, it is broad-spectrum across all three coronaviruses, clinical-stage, and literature-supported (Borgo et al. 2021), but our L1000 test (§3.9) found no transcriptomic reversal of the infection signature, so its support rests on the interactome topology and prior literature rather than on transcriptomic evidence. Every drug link in this study is a database-anchored hypothesis and a starting point for target-validation experiments, not evidence that the drug treats COVID-19.

4.5 The conserved core points to the replication complex

The 42-protein pan-coronavirus core is dominated by targets of the viral replication machinery (Nsp7/8/13), and those baits are the most conserved parts of the virus (§3.10). This is a coherent and encouraging picture for broad-spectrum strategy: the host factors shared across SARS-CoV-2, SARS-CoV-1 and MERS are engaged through the viral proteins least likely to escape by mutation, so a therapy directed at such a host factor would be buffered against both cross-species variation and within-species drift. The caveat is that conservation of the viral bait is a proxy for durability, not proof: a fixed substitution could re-wire a host interaction without abolishing the protein, and our conservation readout covers SARS-CoV-2 Omicron variation rather than the full zoonotic coronavirus range.

4.6 Limitations

The foundational limitation is inherited from the source: an AP-MS map is an incomplete, detection-biased sample that favors abundant, stable complexes (Menche et al. 2015), so both the enrichment results and the missing-component predictions are conditioned on what the screen could see. The comparative analysis is asymmetric — the SARS-CoV-1 and MERS maps come from a dedicated comparative study (Gordon et al. 2020) while the SARS-CoV-2 map is the original Nature map, generated under somewhat different conditions. The disease-context layers each carry their own bounds: the infection and drug-reversal signatures come from single cell lines and timepoints; tissue expression is a proxy for functional dependence, not a demonstration of it; and the L1000 platform measures ~1,000 landmark genes in cancer lines rather than airway epithelium. Every predicted edge needs co-IP validation, every enrichment is a statement about annotation overlap, and every drug-target link is a database record rather than a demonstrated effect on viral replication.

5. Conclusion

Treating the finished SARS-CoV-2 interactome as an object to be interrogated — rather than rebuilt, yields a consistent picture under explicit standards. The nuclear pore is the most cohesively central hijacked complex (confirmed against a degree-matched null); mitochondrial import, the centrosome/PKA machinery and the CUL2 ligase are the most specifically enriched; link prediction nominates testable missing nucleoporins with cross-module skill; and 42 host factors are shared across three coronaviruses, engaged through their most conserved viral proteins. The virus captures these machines post-transcriptionally, in cells it genuinely infects. The central therapeutic tension is that the network's most important and most conserved nodes are undruggable scaffolds, while its approved chemistry sits on peripheral enzymes, a gap that argues for degrader modalities and for prioritizing the conserved, replication-complex-bound core. All results are hypotheses for experimental follow-up; each points to a specific next experiment — a co-IP, a knockdown, an antiviral assay.

Data and code availability

The SARS-CoV-2, SARS-CoV-1 and MERS interaction maps are on NDEX (UUIDs in Methods). STRING v12.0, CORUM 5.3, Reactome, GO, DGIdb, Open Targets, ChEMBL, GTEx v8, the Human Protein Atlas, CELLxGENE, GEO GSE147507, SigCom LINCS and CoV-Spectrum are public. All derived tables (network, centrality, enrichment, link-prediction, drug classification, cross-viral core, infection directionality, tissue context, drug reversal, variant conservation, degree-preserving randomization, stratified link prediction) and all figure-generating code are provided as artifacts accompanying this report.

Appendix A. Reproducibility

Random seeds. All stochastic procedures used fixed seeds. The link-prediction train/test split, the Adamic-Adar and node2vec runs, and the leave-one-out control used seed 42. The degree-preserving randomization used 1,000 independent double-edge-swap rewirings seeded deterministically; the DESeq2 and connectivity analyses are deterministic given their inputs.

Software versions. Python 3.13.14 with networkx 3.6.1, python-igraph 1.0.0, pandas 3.0.3, numpy 2.5.1, scipy 1.18.0, scikit-learn 1.9.0, gensim 4.4.0 (node2vec via DeepWalk-style random walks), statsmodels 0.14.6 (Benjamini-Hochberg correction, cross-checked against a hand-rolled implementation), goatools 1.6.5 (GO enrichment), pydeseq2 0.5.4 (infection differential

expression), `ndex2 3.11.0` (network retrieval). The PDF was rendered with `markdown-it-py 4.2.0` and `weasyprint 69.0`.

Data versions. SARS-CoV-2 map: NDEx `5d97a04a-6fab-11ea-bfdc-0ac135e8bacf` (Gordon et al. 2020, Nature). SARS-CoV-1 and MERS maps: NDEx `ba54a8e5-...` and `c14b148d-...` (Gordon et al. 2020, Science). STRING v12.0 (physical subnetwork, combined score ≥ 700). CORUM 5.3. Reactome and GO retrieved with the goatools-bundled `go-basic.obo` and `goa_human.gaf`. GTEx v8. Human Protein Atlas. GEO GSE147507. SigCom LINCS `11000_cp` library. CoV-Spectrum LAPIS `sars_cov-2_nextstrain_open` dataset. Exact query dates, dataset versions and request identifiers are recorded in the provenance fields of the corresponding data tables.

Artifacts. Every figure is accompanied by the table it was drawn from, and every analysis by the code that produced it, provided as artifacts alongside this report. All literature DOIs resolve against CrossRef/OpenAlex.

Appendix B. Analysis workflow

This study was executed as an agentic analysis pipeline: a controller decomposed the task into independent tracks (network construction and centrality, enrichment, link prediction, drug mapping, cross-viral intersection, and the disease-context and external-validation analyses) and dispatched them as parallel sub-analyses, each returning provenance-tagged tables and figures that were then synthesized into this manuscript under a fixed set of standards (held-out scoring for predictions, multiple-testing correction for enrichment, degree-matched nulls for centrality, and source-level provenance for every value). The design goal of that structure is auditability: because each claim traces to a specific table, database record or accession, the analysis can be re-run or independently checked claim by claim. The scientific conclusions rest on the public data and the methods described in Section 2, not on the orchestration; the workflow is reported here for transparency.

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